Ramzy Rafla

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Education

ETH Zürich *Sept 2026 – Jun 2028*
MSc in Robotics, Systems and Control

EPFL *Sept 2023 – Jul 2026*
BSc in Electrical and Electronic Engineering
- **GPA:** 5.20/6
- **Coursework:** Analysis I–IV, Matrix & Numerical Analysis, Probability & Statistics, Machine Learning & Data Science, Signal Processing, Control Systems, Embedded Systems, Telecommunications, Microwave & Antennas.
- **Teaching Assistant:** Analysis I & III, Introduction to Electricity (2 semesters).

Imperial College London *Summer 2022*
Physics and Mathematics Summer Programme

Experience

Data Science Intern *Cairo, Egypt*
MNT-Halan (Egypt's First Fintech Unicorn) *Jul 2026 – Aug 2026*
- Built a content-based recommendation engine combining customer affinity and product popularity.
- Engineered a 63-feature credit-risk dataset from 300k+ loan applications for default prediction.
- Built a hybrid RAG pipeline (Dense + BM25 + Cross-Encoder), achieving 88% Acc@1 and 100% Acc@4.

Medical AI Research Intern *Martigny, Switzerland*
Idiap Research Institute *Feb 2026 – Jul 2026*
- Developed a 3D CNN pipeline for renal artery abnormality detection in CTA scans.
- Rebuilt the preprocessing workflow to eliminate patient leakage and preserve anatomical geometry.
- Evaluated five image representations, achieving a test AUROC of 0.840.

Artificial Intelligence Engineer *Lausanne, Switzerland*
EPFL Innovation Park *Jul 2025 – Nov 2025*
- Co-developed Corporate Trust Index, an AI system for detecting corporate fraud from weak public signals.
- Built web-scale data pipelines and supported model development; selected for the EPFL Startup Launchpad.

Founding Engineer *Lausanne, Switzerland*
EHL Innovation Hub *Nov 2025 – Jan 2026*
- Built Suite Assist, an AI concierge answering guest questions from hotel documentation.
- Developed the RAG backend using OpenAI embeddings, Pinecone, OpenRouter, and custom guardrails.

Project Manager & Strategic Partnership *Lausanne, Switzerland*
Junior Entreprise EPFL *Feb 2025 – Jul 2026*
- Managed a 100'000 CHF+ portfolio of AI/software client projects while creating partnership within the EPFL tech ecosystem.

Full Stack Engineering Intern *Cairo, Egypt*
Microsoft *Jul 2024 – Aug 2024*
- Built an internal ride-sharing platform using React, ASP.NET Core, and C# to reduce employee commuting emissions.

Projects

Software-Defined FM Stereo Receiver *GNU Radio, DSP, FIR Filters*
- Implemented a complete FM stereo receiver with demodulation, 19 kHz pilot recovery, stereo reconstruction, filtering, and 250-to-48 kHz resampling; validated live reception and implemented an arctangent-free FM demodulator.

Deep Learning Image Classifier *Python, PyTorch*
- Built an end-to-end beverage classifier by scraping ≈ 2,000 images, fine-tuning ResNet18 with augmentations, and implementing a modular training pipeline with checkpointing.

Embedded Game Development on Nintendo DS *C, ARM, libnds*
- Built a real-time game on ARM9/ARM7 processors, implementing timers, interrupts, graphics, audio, touchscreen input, and sprite management directly on the hardware.

FawzyChat *React, FastAPI, OpenRouter*
- Built a multilingual voice assistant supporting transcription, translation, email drafting, custom Arabizi normalization, and real-time browser recording.

Research & Publications

- **Bachelor Thesis:** *Computer-Aided Detection of Renal-Artery Abnormalities in CT Angiography*, EPFL & Idiap Research Institute.

Skills

- **Languages:** English (C2), French (C2), Arabic (Native), Spanish (A2)
- **Programming:** Python, C++, C, TypeScript
- **Frameworks/Libraries:** PyTorch, React, FastAPI, scikit-learn, Pandas, NumPy, Matplotlib, Gtkmm